AQ11.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Choose the correct statements about $f(x, y) = x^3 + xy + y^4 - 4$.f(x, y) = x3 + xy + y4 − 4.

- [ ] $(0, 0)$(0, 0) is a critical point of ff.
- [ ] $(0, 0)$(0, 0) is the only critical point of ff in R^2R2.
- [ ] $f_{xx} = 6x$fxx = 6x
- [ ] $f_{yy} = 12x^2$fyy = 12x2
- [ ] $f_{xy} = 1$fxy = 1
- [ ] $f_{yx} = 0$fyx = 0

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 0)$(0, 0) is a critical point of ff.

$f_{xx} = 6x$fxx = 6x

$f_{xy} = 1$fxy = 1

1 point

If $f(x, y) = \tan^{-1}(x/y)$f(x, y) = tan−1(x/y), then which of the following options are correct.

- [ ] $f_x = \frac{y}{(x^2 + y^2)}$fx = (x2+y2)y
- [ ] $f_{yy} = \frac{2xy}{(x^2 + y^2)^2}$fyy = (x2+y2)22xy
- [ ] $f_{yy} = \frac{-2xy}{(x^2 + y^2)^2}$fyy = (x2+y2)2−2xy
- [ ] $f_{yx} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$fyx = (x2+y2)2x2−y2
- [ ] $f_{yx} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$fyx = (x2+y2)2y2−x2

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_x = \frac{y}{(x^2 + y^2)}$fx = (x2+y2)y

$f_{yy} = \frac{2xy}{(x^2 + y^2)^2}$fyy = (x2+y2)22xy

$f_{yx} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$fyx = (x2+y2)2x2−y2

1 point

If $f(x, y) = x^2 - 3xy^3$ f(x, y) = x2 − 3xy3, then the determinant of the Hessian matrix $Hf(x, y)$ Hf(x, y) is

○
$36xy + 81y^4$ 36xy + 81y4

○
00

○
$-36xy - 81y^4$ −36xy − 81y4

○
$36xy - 81y^4$ 36xy − 81y4

No, the answer is incorrect.
Score: 0
Accepted Answers:
$-36xy - 81y^4$ −36xy − 81y4

Level 2

1 point
Choose the correct statements about $f(x, y) = x^2 + y^4$.f(x, y) = x2 + y4.

☐
(0, 0)(0, 0) is a critical point of ff.

☐
Rank of the Hessian matrix $Hf(0, 0)$Hf(0, 0) is 22.

☐
The determinant of the Hessian matrix $Hf(0, 0)$Hf(0, 0) is 00.

☐
The determinant of the Hessian matrix $Hf(0, 0)$Hf(0, 0) is 11.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(0, 0)(0, 0) is a critical point of ff.
The determinant of the Hessian matrix $Hf(0, 0)$Hf(0, 0) is 00.
Number of critical points of the function $f(x, y) = x^3 + y^3 - 3x - 3y$f(x, y) = x3 + y3 − 3x − 3y is
No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4
1 point
1 point
For which of these functions are the mixed partial derivatives the same at (0, 0)(0, 0).

☐
$f(x, y) = 3x^2 + 7y^3 - 2x^2y^3$ f(x, y) = 3x2 + 7y3 − 2x2y3.

☐
$f(x, y) = \{\begin{matrix} 1 & y \neq 0 \\ 0 & y = 0. \end{matrix}$ f(x, y) = {10y = 0y = 0.

☐
$f(x, y) = |x|(y^2 + 1)$f(x, y) = |x|(y2 + 1).

☐
$f(x, y) = ax + by$f(x, y) = ax + by, where aa and bb are two fixed real numbers.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f(x, y) = 3x^2 + 7y^3 - 2x^2 y^3$ f(x, y) = 3x2 + 7y3 − 2x2y3.
$f(x, y) = ax + by$ f(x, y) = ax + by, where aa and bb are two fixed real numbers.

Check Answers
Your score is: 0/6